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Quantum splitting convolutional neural network-based distributed quantum disease detection model

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A R T I C L E I N F O

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A B S T R A C T

Quantum computing leverages the properties of superposition and entanglement to enable high-dimensional feature extraction from medical images, demonstrating its potential to improve the accuracy of disease detection. However, practical applications are still constrained by the current performance bottlenecks of quantum hardware and affected by the excessive entangling gate operations. In this work, we propose a distributed quantum disease detection (DQDD) model using quantum splitting convolutional neural network. The DQDD employs a hybrid quantum convolutional neural network (QCNN) to capture low-dimensional and high-dimensional features in medical images. We introduce the quantum circuit splitting technique to our model, reconstructing an 8-qubit QCNN with 5 qubits, while simultaneously reducing the entangling operations in each sub-circuit. The model comprises two core components, the Information Distillation mechanism designed to extract pivotal features and the Non-saliency Filtering mechanism dedicated to eliminating irrelevant patterns. Both components operate cohesively, facilitating parallel computation across single or multiple quantum computing devices. Evaluations on ISIC2017 and HAM10000 skin cancer datasets demonstrate that our model outperforms existing quantum models in skin cancer detection, doing so with fewer parameters. Superior generalization capability is further validated through cross-domain testing on the MedMNIST dataset.

1. Introduction

With the rapid advancement of machine learning (ML), emerging AI-driven diagnostic systems [1] have not only enhanced the speed and accuracy of clinical feature analysis and diagnosis, but also reduced human errors, thereby providing reliable support for physicians. In machine learning, neural networks constitute a crucial branch that has demonstrated remarkable efficacy in processing complex data sets. Artificial neural networks with multiple hidden layers exhibit robust feature learning capabilities [2], consequently playing a pivotal role in the field of image recognition. However, existing deep learning models, such as convolutional neural networks (CNNs) [3,4], often struggle to capture subtle pathological features in medical images, such as the irregular distribution of melanocytes in skin cancer detection. This limitation may potentially lead to diagnostic delays. Furthermore, when processing large volumes of medical images, CNNs necessitate increasing the number of model parameters and deepening network layers to extract additional features, which inevitably elevates the computational complexity of the network.

Quantum computing has demonstrated advantages in solving challenging problems intractable for classical computers [5]. Within the field of precision medicine, particularly with pathology images at its core, there's a stringent demand for accurate disease state assessment and prognostic prediction. However, the exceptionally rich detail within pathology images, encompassing a vast number of potentially relevant variables, leads to an extremely complex and high-dimensional feature space, along with intricate dependencies, correlations, and patterns among these variables [6]. These complex high-dimensional associative characteristics correspond well with the principles of quantum superposition and entanglement [7,8]. This theoretically suggests that quantum methods possess an inherent advantage in discerning insights from such data. It is precisely within this context that Quantum Machine Learning (QML) emerges. QML aims to combine the formidable capabilities of quantum computing with classical machine learning to process and

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analyze complex data structures in ways that surpass purely classical approaches [9–12].

Inspired by the remarkable success of classical Convolutional Neural Networks (CNNs) in image processing, researchers have proposed Quantum Convolutional Neural Network (QCNN) [13]. CNNs often face limitations in their linear processing and local connectivity when processing the inherently high-dimensional features, complex intercellular correlations, and non-local patterns found in pathology images [14]. QCNNs address these limitations by encoding data into quantum states and leveraging quantum gate operations to capture exponentially complex relationships between features [15]. This approach allows QCNNs to more effectively explore and learn non-linear, non-local dependencies within the data. This modeling capability, rooted in the principles of quantum mechanics, enables QCNNs to demonstrate stronger feature representation capabilities than classical CNNs in capturing high-dimensional data characteristics, intricate cellular structures, and tissue patterns, making them a powerful tool for analyzing complex pathology images [16–19]. However, to fully exploit the advantages of quantum computing in real-world applications, the noisy intermediate-scale quantum (NISQ) era imposes stringent requirements on both the quantity and quality of qubits.

Unfortunately, the rapid advancement of quantum algorithms and the growing complexity of models have led to quantum computing resource requirements that exceed the capabilities of existing quantum hardware [20]. Additionally, issues such as the neglect of critical features due to excessive entanglement operations [21] persist in QNNs. In the face of hardware limitations in logical qubit counts and the trend toward model simplification, distributed quantum computing [22] has emerged as a critical strategy to achieve quantum advantage and address qubit scalability challenges. This technology aims to expand the number of qubits by integrating multiple quantum network submodules. However, research on distributed quantum neural networks remains notably understudied, largely confined to theoretical frameworks, and necessitates further investigation for practical implementation.

This study proposes a distributed quantum disease detection model (DQDD), which enhances the efficiency and accuracy of medical image classification under quantum resource constraints by harnessing the strengths of both classical and quantum computing. While the concept of quantum circuit splitting has been explored in prior work, our research pioneers its specific application and integration into QCNN for the challenging task of disease detection. This advancement provides a practical approach for deploying more complex quantum models in the NISQ era. The main contributions of the DQDD can be summarized as follows:

- We propose a novel DQDD model that applies quantum circuit splitting techniques to QCNN for disease detection, enabling distributed and comprehensive extraction of both high-dimensional and low-dimensional features from medical images. Meanwhile, the qubit requirements and entangling gate operations in each sub-circuit are reduced by nearly half.
- The proposed model integrates two key components: an Information Distillation mechanism that extracts critical high-dimensional features and a Non-saliency Filtering mechanism that removes irrelevant patterns. This combination enables parallel computation across quantum devices while enhancing the model's discriminative capabilities for disease detection.
- Comprehensive experiments demonstrate the effectiveness of our proposed DQDD model. On the HAM10000 and ISIC2017 skin cancer datasets, compared to existing quantum models, our model achieves state-of-the-art accuracy while utilizing nearly one-tenth of the parameters. Furthermore, its generalizability is validated through evaluation on the MedMNIST dataset.

The structure of this paper is organized as follows: related work is reviewed in Section 2. A detailed description of the model architecture is provided in Section 3. An evaluation of the performance of the model and comparisons with other networks is presented in Section 4. Finally, the paper is concluded and future directions are discussed in Section 5.

2. Related works

The following subsections describe recent advances in CNNs and QCNNs, their applications in medical imaging, and developments in distributed quantum computing technologies.

2.1. CNN for medical image classification

Medical image classification determines the presence and type of diseases by identifying pathological structures within the images, providing a global basis for disease detection. Convolutional neural networks, as a significant branch of deep learning, are utilized in medical image processing. CNN models such as GoogleNet [23], AlexNet [24], and Inception [25] demonstrate outstanding performance in image classification and recognition tasks. In 2018, features were extracted using a pre-trained AlexNet and combined with an ECOC SVM classifier for skin cancer classification by Dorj et al. [26]. In 2021, an automated network for brain tumor classification using SVM and CNN was proposed by Deepak and Ameer [27]. In 2023, a method that combines GoogleNet with the Dynamic Dipper Throated Optimization Algorithm (DDTPSO) for breast cancer classification was employed by Alhussan et al. [28].

2.2. Quantum convolutional neural network

Recent successes in classical neural networks, coupled with advancements in quantum computing, have inspired the investigation of interactions between these technologies, culminating in the development of quantum neural networks [29]. The idea of quantum neural computation was first introduced by Kak [30]. In 2019, a physically realizable quantum convolutional neural network was introduced by Cong et al. [13], resembling the structure of convolutional neural networks, with network layers constructed from parameterized quantum circuits. QCNNs were employed by Qu et al. [31] to extract features from medical images and integrate them with other modal features. In 2024, a custom convolutional neural network deep learning model was proposed by Rao et al. [32], built on a quantum variational circuit with encoding, entanglement, and measurement capabilities.

Hybrid models have been proposed by Liu et al. [33], building on the strengths of both classical neural networks and quantum neural networks. In the work by Liang et al. [34], a hybrid quantum-classical convolutional neural network is presented, which integrates residual block structures with quantum convolutional neural networks. In 2023, a hybrid quantum convolutional neural network was constructed by Qu et al. [35] to extract temporal features from electrocardiogram signals for detecting abnormal heartbeats. In 2024, Olvera et al. [36] proposed an end-to-end hybrid classical-quantum network, where a classical CNN efficiently extracts classical features from EEG signals, followed by a quantum classifier performing higher-complexity classification decisions. By embedding quantum convolutional layers into ResNet50 to construct a hybrid architecture, Lopez et al. [37] validated the effectiveness of hybrid quantum model transfer learning for X-ray-based COVID-19 diagnosis.

Despite the rapid development of QCNNs, their application in medical image analysis remains limited. Therefore, designing a QCNN tailored for medical image classification is crucial.

2.3. Distributed quantum computing

Quantum computing for medical image classification is typically constrained by entanglement gate operations and qubit limitations due to hardware restrictions. A promising solution to this challenge is distributed quantum computing (DQC), which offers greater feasibility and flexibility. In 1993, DQC based on quantum teleportation was proposed by Bennett et al. [38]. In 2000, fault-tolerant quantum logic gates, were constructed by Zhou et al. [39] from Stanford University using a method similar to quantum teleportation.

Among these, quantum circuit segmentation is favored due to its ability to more accurately reconstruct quantum states and its broader applicability. In 2020, quantum circuit splitting was proposed and demonstrated by Peng et al. [40]. In 2022, quantum circuit splitting was utilized by Eddins et al. [41] to express a 2 n-qubit quantum simulation as a linear combination of two n-qubit quantum simulations, effectively doubling the problem size that the chip can handle. In 2024, as described by Harrow and Lowe [42], by employing quantum circuit splitting, involving the measurement of the existing quantum state and re-preparation based on the measurement results, it facilitates the decomposition of the quantum circuit.

3. Methodology

This section delineates the workflow and architecture of the proposed DQDD model. The architecture comprises three functionally coupled modules: a classical feature extraction, the core distributed quantum splitting convolutional network and a task-specific fully connected classifier.

3.1. Workflow

Taking skin cancer detection as a case study, Fig. 1 illustrates the triphasic workflow of our model: (1) Data Acquisition, (2) Feature Processing, and (3) Distributed Quantum Computing & Diagnosis.

Data Acquisition: The medical images utilized for skin cancer diagnosis are predominantly acquired through dermoscopy [43], a non-invasive imaging modality. By combining optical magnification with polarized light filtration, dermoscopy produces high-resolution digital images that reveal subtle microstructures of the skin surface and pathological features in the epidermal layer. These high-information-content images provide valuable datasets for subsequent deep learning-assisted diagnosis, aiding in the improvement of skin cancer diagnostic accuracy [44].

Feature Processing: Feature Processing is divided into data pre-processing and feature extraction. In data pre-processing, we perform the dataset division and data augmentation. The trained model is obtained using the training data, and the performance is evaluated on the testing data based on the classification predictions. The feature extraction module utilizes CNN within the proposed structure to distill low-dimensional features from skin cancer images while reducing the dimensionality of extracted feature vectors, thus adapting them to the dimensional requirements for subsequent integration into quantum networks. In our experiment, the CNN compresses input images into an 8-dimensional feature vector. This vector is encoded into an 8-qubit quantum state, thereby satisfying the input requirements of the subsequent quantum circuit.

Distributed Quantum Computing & Diagnosis: In the distributed quantum computing phase, QCNN is leveraged to extract high-dimensional features from skin cancer images. By introducing quantum circuit splitting technique, a large-scale QCNN circuit is divided into multiple smaller sub-circuits, enabling parallel processing across one or multiple quantum computers. This strategy reduces the entangling gate operations in each sub-circuit, preventing the QCNN from getting trapped in unintended global entanglement, while effectively alleviating hardware constraints. Information Distillation mechanism is designed to identify and amplify discriminative features critical for lesion classification, while Non-saliency Filtering mechanism is engineered to suppress extraneous details that may interfere with lesion recognition. The measurement outcomes from these sub-circuits are fused to generate classification predictions, providing interpretable decision support for medical practitioners to enhance diagnostic accuracy in skin cancer images. In our experiment, within the distributed quantum computing, the final 5 measurement qubits yielded 2^5-dimensional feature values. These values were then used as the input for a fully connected layer, with a softmax function applied for the multiclass classification task.

The architecture of the proposed DQDD is illustrated in Fig. 2, comprising three core modules: a feature extraction module, a distributed quantum splitting convolutional neural network, and a fully-connected classification layer.

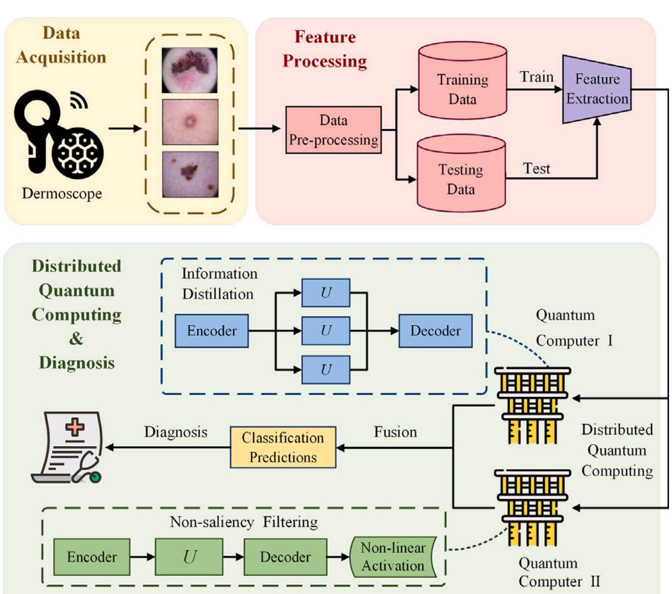

Fig. 1. The workflow of our DQDD model.

3.2. Feature extraction

A lightweight CNN architecture, MobileNetV2 [45], was employed to leverage its ability to capture spatial hierarchical features in images for medical image low-dimensional feature extraction. This model was selected based on the feature extraction performance of the algorithms reported in previous studies involving medical images [46]. This network introduces “Inverted Residuals and Linear Bottlenecks,” which implement an inverse residual structure opposite to that of traditional residual networks. Additionally, the concept of linear bottlenecks is incorporated into MobileNetV2, using linear transformations instead of non-linear activation functions within the residual blocks. This approach significantly reduces computational demands and memory usage, further enhancing the model’s efficiency.

MobileNetV2 initially features a stride-2 convolutional layer for downsampling, followed by seven stages (18 layers total) of inverted residual blocks. Each stage’s parameters include an expansion factor, output channels, repetition count, and stride. At the network’s end, a 1x1 convolution expands features to 1280 channels, which are then processed by global average pooling. The classifier outputs eight feature values, serving as the input for the subsequent quantum network.

3.3. Backbone

To enhance the learning task, we employ a QCNN to extract high-dimensional features. The QCNN can be described using a quantum circuit. These circuits are composed of three main components: quantum state encoding, quantum entanglement layers, and quantum measurement.

We use an 8-qubit QCNN in this study. For the quantum encoding layer, we employ angle encoding, where the 8-dimensional feature values from the upstream module are mapped one-to-one to the 8 qubits and encoded via quantum rotation gates. The QCNN comprises a single convolutional layer constructed using entanglement blocks. Specifically,

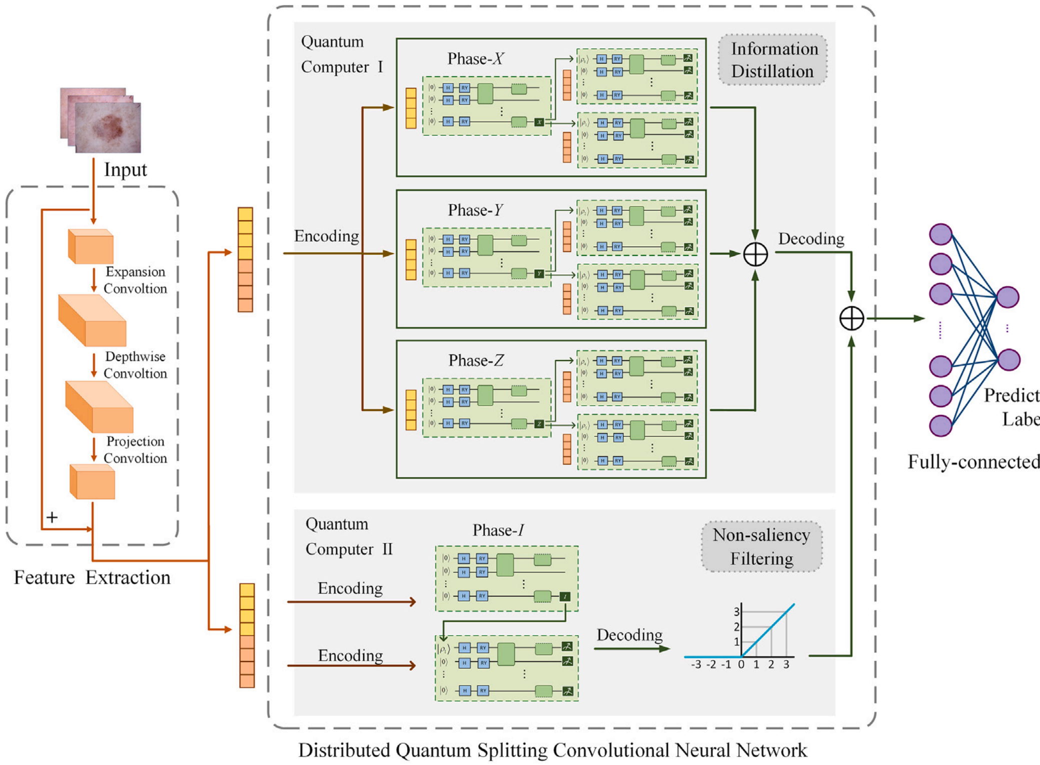

Fig. 2. Architecture of the proposed DQDD.

adjacent qubits are entangled, and additionally, the 1st qubit is entangled with the 4th, and the 4th with the 8th. Each entanglement block consists of 6 quantum gates, resulting in a total of 9 such blocks (54 quantum rotation gates in total) within the entire entanglement layer. Finally, we perform *Pauli-Z* measurements. The specific architecture is detailed further below.

Our research employs angle encoding to embed classical data points $D \subset \mathbb{R}^n$ into quantum states. For any subset $D' = (x_0, \dots, x_{n-1}) \subseteq D$, an injective map $f : D' \to \mathcal{H}^m$ is defined as:

$$f(D') = \frac{1}{C}\sum_{i=1}^{n} x_i \,|i\rangle, \quad \text{where } C = \sqrt{\sum\nolimits_{i=1}^{n} x_i^2} \tag{1}$$

Here, $|i\rangle$ denotes the computational basis states of the n-qubit Hilbert space $\mathcal{H}^n$. This encoding is implemented via parameterized quantum circuits using the following gates:

$$H = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \quad RY(\theta) = \begin{bmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{bmatrix} \tag{2}$$

The H-gate initializes superposition states, while $RY(\theta)$-gates perform data-dependent rotations on qubits.

The transformation of classical information x_i into a quantum state is achieved via a single-qubit rotation gate R, as shown in Eq. (3)

$$\left|x_i\right\rangle = \otimes_{i=0}^{N} R(x_i) \left|0^N\right\rangle \tag{3}$$

The quantum entanglement layer also employs single-qubit $RZ(\theta)$-gates and two-qubit controlled gates $CNOT$-gate:

$$RZ(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}, \quad CNOT = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix} \tag{4}$$

For controlled-RY operations, the $CRZ(\theta)$-gate representation is:

$$CRY(\theta) = \begin{bmatrix} I & 0 \\ 0 & RY(\theta) \end{bmatrix} \tag{5}$$

Control qubits conditionally apply operations to target qubits via these gates.

QCNNs inherit the hierarchical architecture of classical CNNs through quantum convolutional and pooling layers. The quantum convolutional layer implements feature extraction by utilizing a quasi-local unitary operator U that is applied with translational invariance, and it employs a matrix product states (MPS) structure with bond dimension D. This structure allows for processing and feature extraction from the quantum state with limited depth, similar to the convolution operation in classical CNNs, enabling $O(D^2)$ complexity compression of d^n-dimensional quantum states.

The MPS architecture is characterized by a ladder-like configuration that decomposes high-dimensional tensors into a series of lower-dimensional tensor products. These lower-rank tensors are interconnected through internal indices, referred to as the bond dimension D. The advantage of MPS lies in its ability to approximate complex quantum states with reduced computational complexity. Fig. 3 illustrates the quantum circuit designed in this study. Each U in the MPS circuit is substituted with a simple variational quantum circuit, which represents the convolutional kernel, as illustrated in Fig. 4. This circuit can be simulated, implemented on hardware, and optimized like other variational quantum circuits.

The quantum convolutional layer significantly compresses the feature space through its translational invariance and entanglement operations, rendering quantum pooling layers unnecessary.

The quantum state is converted into a classical state through measurement, with the results returned as a flat array containing the probabilities of measuring the computational basis states in the current configuration. If the measurement is constrained to N measurement circuits, the marginal probabilities can be obtained, resulting in a returned

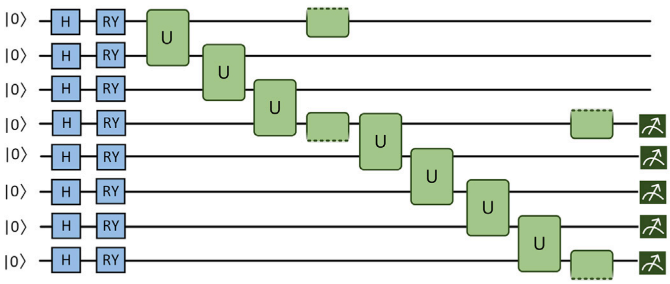

Fig. 3. Full QCNN architecture prior to quantum circuit splitting optimization.

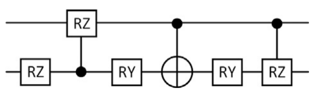

Fig. 4. Quantum convolutional kernel: the use and distribution of specific entanglement gates in each entanglement block.

array size of 2^n:

$$P = (|\langle i_0\ |\psi\rangle|^2, |\langle i_1\ |\psi\rangle|^2, \cdots, |\langle i_{2^n-1}\ |\psi\rangle|^2) \tag{6}$$

where $|\psi\rangle$ is the current quantum state of the system, and $|i_k\rangle$ is a computational basis state.

3.4. Distributed quantum splitting convolutional neural network

Current quantum computing hardware is constrained by a limited number of qubits, making it challenging to process large-scale quantum circuits, especially when the circuit size exceeds the hardware capacity. Additionally, to more effectively extract high-dimensional features from medical images, we must introduce more complex entangling gate operations. However, the addition of excessive entangling gates may cause the overall network to become trapped in unintended global entanglement, leading to a degradation in the ability to express local features of quantum states and weakening the ability to capture key patterns in the data due to strong correlations among qubits.

By employing quantum circuit splitting technique, large circuits can be decomposed into a series of smaller circuits, thereby reducing the entangling gate operations in each sub-circuit. This approach ensures the transmission of key information while increasing the independence among qubits, thus enhancing the ability to express local features of quantum states. Furthermore, this method enables parallel execution of sub-tasks on one or more quantum processors, allowing for more complex computational tasks to be performed on existing hardware.

3.4.1. Quantum circuit splitting

In a quantum circuit, any quantum operation represented by a unitary matrix U can be decomposed into a combination of orthogonal matrices formed by the Pauli basis $\{I, X, Y, Z\}$.

$$U = \frac{Tr(UI)I + Tr(UX)X + Tr(UY)Y + Tr(UZ)Z}{2} \tag{7}$$

Based on the following spectral decomposition relationship, the Pauli basis can be rewritten as:

$$\begin{aligned} I &= |0\rangle\langle 0| + |1\rangle\langle 1| \qquad & X &= |+\rangle\langle +| - |-\rangle\langle -| \\ Y &= |+i\rangle\langle +i| - |-i\rangle\langle -i| \qquad & Z &= |0\rangle\langle 0| - |1\rangle\langle 1| \end{aligned} \tag{8}$$

The quantum state ρ can be split as a linear combination of the eigenstates of the observables (Pauli matrices). The coefficients in this expansion are governed by the expectation values of ρ with respect to the corresponding observables, expressed through the trace operation:

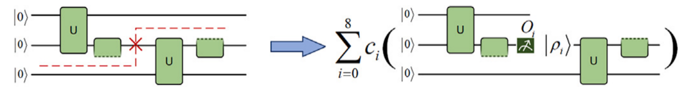

Fig. 5. Diagram of the circuit using the quantum circuit splitting technique.

$$\rho = \frac{1}{2}\Big(\mathrm{Tr}(\rho I)I + \mathrm{Tr}(\rho X)X + \mathrm{Tr}(\rho Y)Y + \mathrm{Tr}(\rho Z)Z\Big) \tag{9}$$

Therefore, for the split channel, it is composed of measurements of the Pauli observables O_i and preparations of the corresponding eigenstates ρ_i.

Employing the four orthogonal Pauli bases $\{X, Y, Z, I\}$, the measurement and preparation operations for the decomposed quantum circuit are implemented. Assuming $2c_i$ signifies the associated eigenvalues. The complete correspondence between Pauli observables, eigenstates, and eigenvalues is as follows:

$$\begin{aligned}
&O_0 = X,\ \rho_0 = |+\rangle\langle +|, && c_0 = +1/2; && O_1 = X,\ \rho_1 = |-\rangle\langle -|, && c_1 = -1/2;\\
&O_2 = Y,\ \rho_2 = |+i\rangle\langle +i|, && c_2 = +1/2; && O_3 = Y,\ \rho_3 = |-i\rangle\langle -i|, && c_3 = -1/2;\\
&O_4 = Z,\ \rho_4 = |0\rangle\langle 0|, && c_4 = +1/2; && O_5 = Z,\ \rho_5 = |1\rangle\langle 1|, && c_5 = -1/2;\\
&O_6 = I,\ \rho_6 = |0\rangle\langle 0|, && c_6 = +1/2; && O_7 = I,\ \rho_7 = |1\rangle\langle 1|, && c_7 = +1/2.
\end{aligned} \tag{10}$$

The original quantum circuit requires a combination of 8 distinct quantum channels to implement the identity mapping corresponding to Eq. (9).

$$\rho = \sum_{i=0}^{8} c_i \mathrm{Tr}(\rho O_i)\rho_i \tag{11}$$

Our experiment utilizes quantum circuit splitting technique to bisect the original QCNN quantum circuit. As shown in Eq. (10), the partitioned circuit is divided into four distinct subgroups, each containing an upper subcircuit and two lower subcircuits. Via combination of the measurement outcomes from these subgroups (as shown in Eq. 11), the original QCNN circuit achieves high-fidelity reconstruction. Fig. 5 shows the diagram of the circuit using the quantum circuit splitting technique. The Pauli matrices span the observable space in the complex Hilbert space, thereby enabling high-dimensional interpretation of medical imaging details, including texture, color, and scale. Meanwhile, each sub-circuit retains half of the entangling gates from the original circuit, increasing qubit independence and preventing the QCNN from becoming trapped in unintended global entanglement.

3.4.2. Information distillation

Information Distillation mechanism introduces the quantum circuit splitting technique into the QCNN framework proposed in this study. To maximize the topological advantages of this technique for quantum resource efficiency, the circuit segmentation is specifically implemented at the midpoint of the fourth qubit in the 8-qubit QCNN.

As shown in Eq. (10), measurements in the I or Z basis physically correspond to equivalent quantum circuit realizations. Consequently, the Pauli basis $\{X, Y, Z\}$ is sufficient for enabling QCNN to achieve high-dimensional feature extraction. Based on the quantum operator decomposition theorem, the original circuit is reconstructed into three cascade groups: each group comprises a 4-qubit upper sub-circuit for high-dimensional feature extraction of the upper input segment, along with two 5-qubit lower sub-circuits. The lower sub-circuits integrate one memory qubit from the upper sub-circuit and 4 qubits dedicated to high-dimensional feature extraction of the lower input segment, forming a total of nine independently configurable quantum sub-circuit systems.

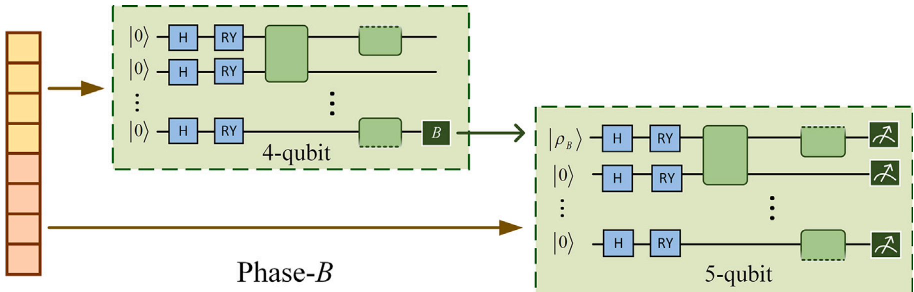

Fig. 6. The circuit of a single Phase-B.

The schematic of the Phase-B circuit, designed using Pauli basis B (Pauli basis X, Y, Z), is illustrated in Fig. 6.

Since the splitting is applied at the midpoint of the original QCNN, each sub-circuit employs only half the number of entanglement gates compared to the original circuit. This design enhances diagnosis-relevant local entanglement such as pathology-specific feature interactions.

The mechanism significantly enhances the recognition of key pathological biomarkers in disease detection, such as the identification of lesion boundary irregularities, chromatic heterogeneity, and morphological deviations in tissue microstructure in skin cancer, all of which are critical factors for malignancy detection.

3.4.3. Non-saliency filtering

Due to the superposition property of quantum systems, the state of a qubit is a linear combination of basis states:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1. \tag{12}$$

To ensure that all measurement probabilities sum to 1 while satisfying Eq. (11), we decompose and reconstruct the I operator, and embed the Pauli-I “measurement-preparation” within the QCNN at the position specified in the “Information Distillation” section, thus generating the Phase-I set. By combining the results of Phase-$\{X, Y, Z, I\}$, we can fully reconstruct the original QCNN used in this work.

In medical imaging (e.g., dermatoscopic diagnosis), subtle features often encode critical pathological information. To enhance the discriminative capability of the proposed distributed quantum splitting convolutional neural network for medical imaging while behaving the distinction between Phase-Z and Phase-I in extracting high-dimensional features, we propose a Non-saliency Filtering mechanism. This mechanism applies a nonlinear transformation to the output vector of Phase-I, eliminating non-positive high-dimensional features and enabling the measurement observables to focus on key pathological characteristics. The structure of the Non-saliency Filtering mechanism is illustrated in Fig. 7.

This mechanism suppresses the model’s fitting to non-critical attributes (e.g., subject-specific skin texture, skin color, and imaging perspective biases) while retaining disease-relevant features, thereby reducing detection error rates caused by spurious correlations.

4. Experiments

In this section, we detail the experimental setup, present the model’s performance on the datasets, and demonstrate the model’s efficacy through comparative and ablation studies.

4.1. Datasets and evaluation metrics

To assess the diagnostic performance of our model in disease detection, we adopted two publicly accessible dermoscopy medical image collections: the HAM10000 dataset [47] and ISIC2017 dataset [48]. To further evaluate its generalizability across diverse medical imaging modalities, we included a cross-domain X-Ray dataset, MedMNIST dataset [49]. The first and second datasets are large-scale dermoscopic image multiclass datasets released by the International Skin Imaging Collaboration (ISIC) for the ISIC 2018 and 2017 competitions. The third dataset is the pneumonia dataset from MedMNIST, which consists of chest X-ray images for binary classification.

To address the imbalance in dermoscopic images, we applied oversampling and undersampling techniques, resulting in 1372 and 2000 images for each class, respectively . Additionally, data of varying sizes were normalized and cropped to a uniform size of 224×224 pixels to ensure consistency and comparability. The information about the dataset is shown in Table 1. To enhance sample diversity and improve model stability [50], data augmentation was applied to the training dataset. All images were augmented with a 50 % probability of random horizontal flipping. Additionally, brightness and contrast adjustments were applied by randomly scaling within the range [0.8, 1.2].

Multiple evaluation metrics were employed to quantify the performance of the experimental results, including area under curve (AUC), accuracy, recall, precision, F1-score, and specificity [51].

4.2. Experimental setup

In our study, the computational resources consisted comprised of four NVIDIA GeForce RTX 4090 GPUs, a 192-core Intel CPU, and

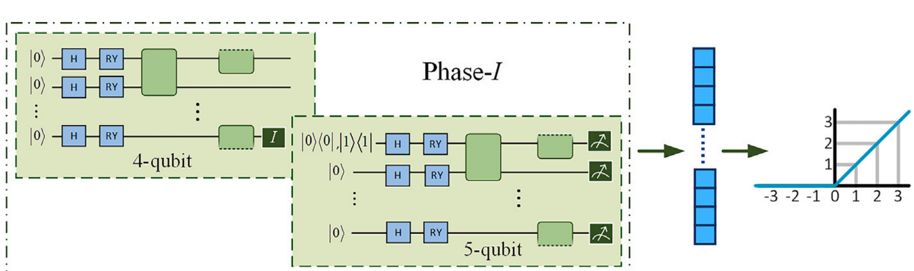

Fig. 7. The structure of the Non-saliency Filtering.

Table 1
Datasets used in the experiments.

Dataset	Categories	Count (before)	Count (after)
HAM10000	AK	327	2000
	BCC	514	
	BKL	1099	
	DF	115	
	MEL	1113	
	NV	6705	
	VASC	142	
ISIC2017	BKL	254	1372
	MEL	374	
	NV	1372	
MedMNIST	NL	1583	1266
	PNA	4273	3418

Table 2
Performance on HAM10000.

Categories	Specificity	Precision	Recall	F1-score
AK	98.65 %	93.10 %	99.54 %	96.21 %
BCC	98.95 %	94.00 %	96.08 %	95.03 %
BKL	97.74 %	86.40 %	83.45 %	84.90 %
DF	**99.92 %**	**99.47 %**	99.47 %	**99.47 %**
MEL	96.49 %	80.05 %	83.42 %	81.70 %
NV	98.10 %	86.23 %	75.59 %	80.56 %
VASC	99.79 %	98.71 %	**100 %**	99.35 %

128 GB of memory. The hybrid quantum model framework was implemented using Python 3.11 with PyTorch 2.2.2 for classical deep learning components and leveraged PennyLane 0.34.0's high-performance lightning.qubit backend for critical quantum simulations.

In addition to the MedMNIST dataset, which is divided into an 8:1:1 ratio, other datasets are typically split into training and testing sets with an 8:2 ratio. During the model training phase, we use an Adadelta optimizer with the learning rate of 0.05. We use cross-entropy loss as shown in Eq. (13), and set the batch size to 16.

$$\mathcal{L}(y, \hat{y}) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{ic}\log(\hat{y}_{ic}) \tag{13}$$

Here, y is a one - hot encoded matrix with a dimension of $[N \times C]$ representing the true labels, and $\hat{y}$ is a probability matrix with a dimension of $[N \times C]$ representing the predicted labels.

4.3. Experimental results

4.3.1. Performance evaluation on the HAM10000

The performances of the DQDD in the 7-class classification on the HAM10000 dataset are presented in Table 2. For all categories, DF (dermatofibroma) performed best, achieving 99.47 % in recall, precision, and F1-score, with 99.92 % specificity. VASC (vascular lesions) achieved a perfect 100 % recall, which is vital for catching all pathological cases in medical settings. Its specificity, precision, and F1-score all exceeded 98 %, making it excellent for high-risk screening. However, NV (nevus) recall was lower at 75.59 %, meaning nearly 25 % of samples were missed, a big risk for melanoma monitoring. As shown in Fig. 8, most misclassifications in the HAM10000 dataset were for MEL (melanoma). This is a critical error, potentially delaying NV monitoring and causing unnecessary anxiety due to melanoma over-diagnosis. Overall, our model achieved 91.14 % accuracy on this dataset, showing strong performance in this 7-class classification task.

4.3.2. Performance evaluation on the ISIC2017

The performances of the DQDD in the 3-class classification on the ISIC2017 dataset are presented in Table 3. Among the categories, the best performance was observed for NV, with a recall of 100 %, precision of 97.31 %, and f1-score of 98.63 %, the highest across all

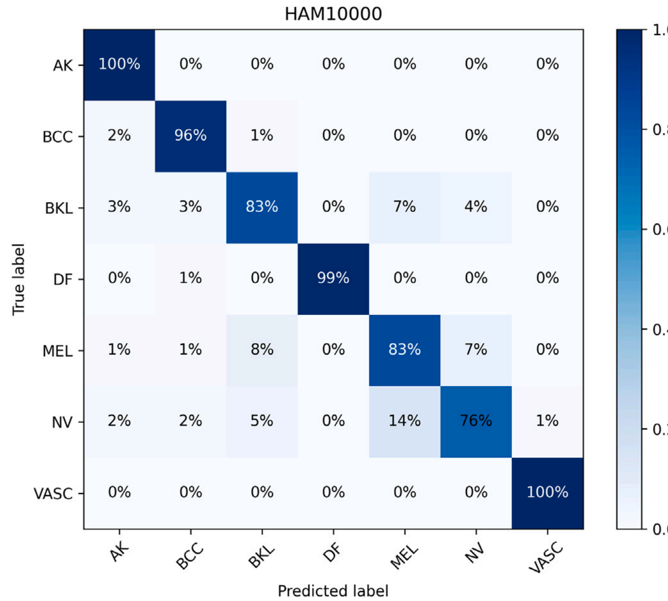

Fig. 8. Confusion matrix showing the performance of DQDD on the HAM10000 dataset. The rows represent the true labels, and the columns represent the predicted labels. The values within the cells indicate the percentage of samples from that class that were predicted as each respective class.

Table 3
Performance on ISIC2017.

Categories	Specificity	Precision	Recall	F1-score
BKL	94.64 %	89.55 %	97.35 %	93.28 %
MEL	**98.73 %**	97.08 %	85.98 %	91.19 %
NV	98.50 %	**97.31 %**	**100 %**	**98.63 %**

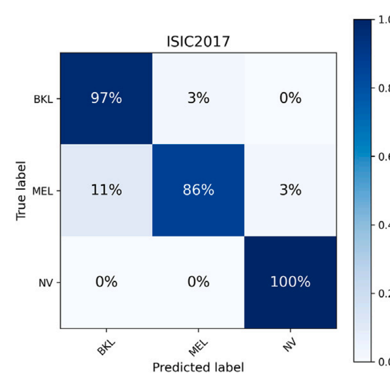

Fig. 9. Confusion matrix showing the performance of DQDD on the ISIC2017 dataset. The rows represent the true labels, and the columns represent the predicted labels. The values within the cells indicate the percentage of samples from that class that were predicted as each respective class.

categories. This indicates exceptionally high sensitivity in detecting nevi, which is a clinically desirable characteristic. In comparison, MEL (melanoma) achieved a specificity of 98.73 %, but recall was relatively lower at 85.98 %. Furthermore, the precision for BKL (seborrheic keratosis) was below 90 %. As shown in the confusion matrix in Fig. 9, the prediction accuracy for MEL was 86 %, with 11 % of the samples misclassified as BKL, representing a potentially life-threatening diagnostic delay. An overall classification accuracy of 94.54 % was achieved. DQDD demonstrates excellent classification performance, in the context

Table 4
Results of 3-fold cross-validation.

Dataset	Accuracy	Specificity	Precision	Recall	F1-score
HAM10000	90.67 %	97.43 %	90.86 %	90.59 %	90.53 %
ISIC2017	93.20 %	96.63 %	93.20 %	93.18 %	93.16 %

Table 5
Performance on MedMNIST.

Categories	Specificity	Precision	Recall	F1-score
NL	**97.94 %**	**96.69 %**	74.79 %	84.34 %
PNA	76.77 %	86.68 %	**98.46 %**	**92.20 %**

Table 6
Benchmark on MedMNIST.

Model	ACC	AUC
ResNet-18	86.43 %	**95.55 %**
auto-sklearn	85.53 %	94.18 %
classical AutoKeras	87.76 %	94.72 %
DQDD	**89.58 %**	94.87 %

of a three-class classification task. However, MEL recall highlights a critical area for improvement, aligning with clinical safety priorities where enhancing category recall metrics is paramount.

To enhance the evaluation of the model's generalization capability, we employed 3-fold cross-validation. As shown in Table 4, the proposed model demonstrates strong generalization performance across both datasets.

4.3.3. Performance evaluation on the MedMNIST

The performances of the DQDD in the binary classification task on the MedMNIST dataset are presented in Table 5. The model achieved a recall of 98.46 % for PNA (Pneumonia), indicating it correctly identified the vast majority of true pneumonia cases. This high sensitivity is crucial for patient safety, as missed pneumonia can lead to severe consequences. In the category NL (normal), the specificity and precision are highest, at 97.94 % and 96.69 %, respectively. However, possibly due to the imbalance in the dataset, NL showed a lower recall (74.79 %) and f1-score (84.34 %), which clinically could lead to potentially unnecessary medication or patient anxiety.

In benchmark experiments, as shown in Table 6, DQDD outperforms the majority of the baselines.

The enhanced detection performance of DQDD underscores its superior discriminative capabilities in dermatological image characterization, attributed to the model's quantum-enhanced capacity to capture non-local morphological patterns often overlooked by classical convolutional architectures. Evaluations on the MedMNIST dataset further demonstrate cross-domain robustness, with DQDD achieving statistically comparable performance metrics in both skin cancer detection and heterogeneous diagnostic tasks, such as X-ray imaging. This confirms its generalizability across clinically distinct imaging domains.

4.3.4. Comparison experiments and analysis

DQDD was compared with existing methods, which include CNNs such as ResNet34 [52], DenseNet121 [53], SA-VGG [54]; Transformer [55]; and QCNNs such as Qst-Cla [56], HQCNN [57], and HQNN-Parallel [58]. To ensure a fair comparison, all experiments were run on the same hardware. We used consistent input data, including the same data augmentation and dataset division. The loss function, learning rate, and training strategy were also identical for all models. This allowed us to directly compare various evaluation metrics on the HAM10000 and ISIC2017 medical image datasets, focusing purely on differences among model architectures. The results demonstrate that our proposed model achieves excellent performance on both datasets.

On the HAM10000 dataset (Table 7), DQDD achieves an accuracy of 91.14 %. This high accuracy reflects the outstanding overall classification performance of the model, as well as its advantage in extracting complex image features. In terms of specificity, DQDD achieves nearly 100 %, indicating exceptional performance in identifying negative class samples. Both precision and recall exceed 91 %, further highlighting the proposed model's stability in recognizing positive class samples. In comparison, other hybrid quantum models such as Qst-Cla and HQCNN did not surpass 90 % in performance metrics except for specificity.

On the ISIC2017 dataset (Table 8), DQDD achieved an accuracy of 94.54 %, significantly outperforming other CNNs such as ResNet34 (90.53 %), DenseNet121 (88.47 %), and SA-VGG (88.59 %). Additionally, it achieved a specificity of 98.50 %, showcasing an exceptional ability to identify negative class samples. Precision, F1-score, and

Table 7
Performance comparison on HAM10000.

Dataset	Model	Accuracy (%)	Specificity (%)	Precision (%)	Recall (%)	F1-score (%)	Para (M)
HAM10000	ResNet34 [52]	89.35	99.71	89.18	89.33	89.14	20.32
	DenseNet121 [53]	89.50	**99.92**	89.44	89.50	89.33	6.72
	SA-VGG [54]	85.53	99.46	86.00	85.42	85.49	45.75
	Transformer [55]	91.12	98.69	91.03	91.06	90.97	6.49
	Qst-Cla [56]	87.92	99.71	88.79	87.67	87.23	78.23
	HQCNN [57]	73.32	98.92	75.07	73.75	72.98	84.01
	HQNN-Parallel [58]	82.39	98.96	83.45	82.32	82.02	11.50
	DQDD	**91.14**	98.73	**91.14**	**91.09**	**91.03**	**2.16**

Table 8
Performance comparison on ISIC2017.

Dataset	Model	Accuracy (%)	Specificity (%)	Precision (%)	Recall (%)	F1-score(%)	Para (M)
ISIC2017	ResNet34 [52]	90.53	96.64	90.38	90.37	90.29	20.32
	DenseNet121 [53]	88.47	93.83	88.76	88.29	87.98	6.71
	SA-VGG [54]	88.59	96.26	88.45	88.49	88.39	45.73
	Transformer [55]	88.38	94.89	88.66	88.24	88.10	6.49
	Qst-Cla [56]	80.46	89.91	81.47	80.12	78.78	78.23
	HQCNN [57]	81.31	96.82	81.22	81.06	81.13	84.01
	HQNN-Parallel [58]	79.24	88.59	79.47	78.90	78.02	11.50
	DQDD	**94.54**	**98.50**	**94.65**	**94.44**	**94.37**	**2.16**

Table 9
The *p*-value was computed using the non-parametric McNemar-Bowker test to compare the proposed model to comparative models.

Model (vs. DQDD)	*p*-value
ResNet34	1.43e-09
DenseNet121	1.27e-07
SA-VGG	1.54e-05
Transformer	1.02e-07
Qst-Cla	1.22e-10
HQCNN	5.04e-04
HQNN-Parallel	1.61e-07

Table 10
Performance comparison under different measurements on HAM10000 and ISIC2017.

Dataset	Pauli Basis	Accuracy	Specificity	Precision	Recall	F1-score
HAM10000	X-basis	89.57 %	98.00 %	89.91 %	89.54 %	89.53 %
	Y-basis	**90.64 %**	**98.43 %**	**90.81 %**	**90.53 %**	**90.46 %**
ISIC2017	X-basis	93.66 %	97.31 %	93.62 %	93.57 %	93.60 %
	Y-basis	**94.41 %**	**98.69 %**	**94.37 %**	**94.29 %**	**94.32 %**

recall all exceeded 94 %, indicating strong identification of positive class samples. In contrast, other quantum models, such as Qst-Cla, HQCNN, and HQNN-Parallel, exhibited accuracies nearly 15 % lower than that of DQDD. This difference may be attributed to the complexity of their methods, indicating that our network structure is both simpler and more effective.

Furthermore, in terms of number of parameters, DQDD contains only 2.16 M parameters, which is significantly smaller than other models. While maintaining outstanding performance, significant model size compression has been achieved, making it especially suitable for resource-limited medical imaging tasks. This demonstrates that excellent performance can still be achieved in resource-constrained environments.

Several classical methods, using the same dataset, have achieved strong performance [59]. These methods leverage well-established image processing techniques, such as Super-Resolution (SR) processing [60], which generates clearer, more detailed high-resolution images from low-resolution skin disease images. The 2D Wavelet Transform [61] is employed to capture multi-resolution local information via signal processing, extracting distinctive features from skin lesions. Furthermore, more optimized learning strategies such as transfer learning and fine-tuning [62] enable faster training of the proposed models on limited, relevant medical image training datasets, allowing for better adaptation to the specific data distribution and features of skin disease images. This offers significant guidance for our future work, extending beyond architectural improvements to the model itself.

To demonstrate the statistical significance of our model against competing models, we used the McNemar-Bowker test to compute the chi-square (χ^2) statistic. As shown in Table 9, our model exhibits significant differences from all comparison models ($p < 0.05$).

The implementation of measurements in different Pauli bases is realized through quantum rotations and unitary transformations. Specifically, measuring *Pauli-X* is equivalent to applying an H-gate to the quantum state, then measuring *Pauli-Z*. Measuring *Pauli-Y* is equivalent to applying an $S^\dagger$-gate followed by an H-gate to the quantum state, then measuring *Pauli-Z*. Table 10 presents results from different measurement bases.

The results obtained using different measurement bases confirm that our QCNN ansatz includes trainable rotation gates about the X, Y, and Z axes. The number of rotation gates for the Y and Z axes is similar and exceeds that for the X-axis. Consequently, when using *Pauli-Y*, the performance metrics are on par with the default *Pauli-Z* results, with fluctuations less than 1 %. However, metrics from *Pauli-X* are slightly lower than those from the default *Pauli-Z*. This indicates that our proposed model captures rich information across all angles, demonstrating its robust quantum feature extraction [63] capabilities.

Table 11
Performance comparison under different numbers of qubits on HAM10000 and ISIC2017.

Dataset	Qubit number	Training time (min)	Accuracy	Specificity	Precision	Recall	F1-score
HAM10000	4	**30**	84.17 %	97.37 %	83.89 %	84.15 %	83.93 %
	8	90	**91.14 %**	**98.73 %**	**91.14 %**	**91.09 %**	**91.03 %**
ISIC2017	4	**15**	87.75 %	92.11 %	79.44 %	91.98 %	83.90 %
	8	40	**94.54 %**	**98.50 %**	**94.65 %**	**94.44 %**	**94.37 %**

As shown in Table 11, all evaluation metrics for the 4-qubit configuration were significantly lower (by 2 %–18 %) than those of the 8-qubit configuration, highlighting the benefits of increased quantum dimensionality for this task. While higher qubit counts theoretically enable richer expressivity, they also lead to exponential growth in training time due to the expanded state space. Furthermore, on current NISQ devices, larger qubit numbers introduce more noise and crosstalk, often degrading practical performance. The 8-qubit configuration emerged as the optimal choice, effectively balancing feature extraction power, training efficiency, and hardware limitations.

4.3.5. Ablation experiments and analysis

To evaluate the impact of the quantum circuit splitting technique on the model, ablation experiments were conducted on both medical datasets, HAM10000 and ISIC2017, as shown in Table 12. After adopting distributed computing based on quantum circuit splitting, DQDD showed an improvement of approximately 2 %–3 % in various metrics. This indicates that the introduction of quantum circuit splitting not only overcomes hardware limitations and reduces the required theoretical quantum bits but also enhances the learning capability of the model. These results strongly demonstrate the feasibility of quantum circuit splitting.

Additionally, we conducted ablation experiments on data augmentation using the same datasets, as shown in Table 13. The results show that the augmentation provided only marginal improvements, suggesting that adopting simple data augmentation approaches has limited effectiveness.

Table 12
Ablation experiments on quantum circuit splitting technique on two datasets.

Dataset	Splitting	Accuracy	Specificity	Precision	Recall	F1-score
HAM10000	✗	89.57 %	99.65 %	89.75 %	89.59 %	89.50 %
	✔	**91.14 %**	**98.73 %**	**91.14 %**	**91.09 %**	**91.03 %**
ISIC2017	✗	91.26 %	98.13 %	91.26 %	91.20 %	91.11 %
	✔	**94.54 %**	**98.50 %**	**94.65 %**	**94.44 %**	**94.37 %**

Table 13
Ablation experiments on data augmentation for two datasets.

Dataset	HAM10000		ISIC2017	
	Without augmentation	Augmentation	Without augmentation	Augmentation
Accuracy	90.17 %	**91.14 %**	93.50 %	**94.54 %**
Specificity	98.10 %	**98.73 %**	97.44 %	**98.50 %**
Precision	90.05 %	**91.14 %**	93.45 %	**94.65 %**
Recall	90.44 %	**91.09 %**	93.64 %	**94.44 %**
F1-score	90.32 %	**91.03 %**	93.70 %	**94.37 %**

5. Conclusions and future work

In this paper, a novel distributed quantum disease detection model is presented, which integrates lightweight CNN with distributed quantum splitting convolutional neural network. The hybrid architecture employs classical computing for extracting low-dimensional features and quantum computing for capturing high-dimensional patterns, while maintaining diagnostic accuracy with low parameter requirements.

A quantum circuit splitting technique is innovatively implemented within the quantum circuit design, enabling 5 qubit simulation of 8-qubit QCNN, while reducing the number of entangling gate operations required for each QCNN. This compression strategy reduces resource demands on noisy intermediate-scale quantum (NISQ) devices while retaining quantum advantages. The distributed quantum splitting convolutional neural network implements two dedicated components: an Information Distillation mechanism that amplifies critical high-dimensional feature details, and a Non-saliency Filtering mechanism that suppresses interference patterns.

Experimental evaluations conducted on two skin cancer datasets (ISIC2017 and HAM10000) and a cross-domain X-Ray pneumonia dataset (MedMNIST) empirically validate the capabilities of the proposed framework. Results demonstrate superior disease detection performance, suitability for the NISQ era, and strong generalizability.

Quantum computing, still in its nascent stages, holds the promise of becoming a powerful tool in healthcare and medicine. However, significant technical hurdles currently limit its widespread application in these fields. We are presently in the NISQ era, characterized by the limited hardware capabilities of existing quantum computers. Physical qubits are highly susceptible to errors due to noise from decoherence and environmental interactions. Furthermore, the strict requirements for quantum computers and the complexity of existing healthcare systems make it challenging to use quantum computing in clinical settings. With continuous advancements in quantum hardware and algorithms, particularly in scalability, error correction, and quantum-classical hybrid systems, we anticipate the emergence of the first quantum-assisted medical applications in the foreseeable future.

Given the critical impact of medical diagnostics on patient safety, our future work will prioritize optimizing existing workflows. We plan to introduce more advanced image preprocessing techniques, such as wavelet transforms and Gaussian filtering, to enhance extraction of clinically relevant features from medical images. Concurrently, we will explore more efficient learning strategies, like transfer learning with pretrained models, improving model characterization capabilities and accelerating convergence.

Beyond that, we aim to optimize our model across broader datasets and diverse tasks. We will leverage quantum technology's unique ability to classify complex multimodal data, like EEG signals and brain X-rays, leveraging quantum-enhanced feature extraction.

Finally, we plan to explore more robust model structures for noisy environments, aiming for real-world applications on quantum hardware. By improving quantum circuit compilation strategies and incorporating noise reduction algorithms, this work will validate our method's scalability and lay the groundwork for practical quantum-AI integration.

CRediT authorship contribution statement

Yangyang Li: Validation, Supervision, Methodology, Conceptualization. **Zhengya Qi:** Writing – review & editing, Writing – original draft, Formal analysis. **Yuelin Li:** Writing – review & editing, Formal analysis. **Haorui Yang:** Writing – review & editing, Data curation. **Ronghua Shang:** Investigation. **Licheng Jiao:** Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available upon request.

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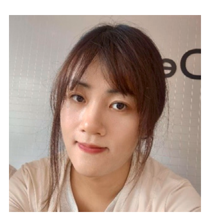

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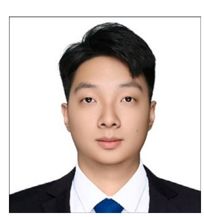

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